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ASP3231 Assignment 3
Semester One 2026
Faculty of Science
School of Physics and Astronomy

AUTHORISED MATERIALS

OPEN BOOK

YES ☒

NO ☐

CALCULATORS

YES ☒

NO ☐

SPECIFICALLY PERMITTED ITEMS

YES ☒

NO ☐

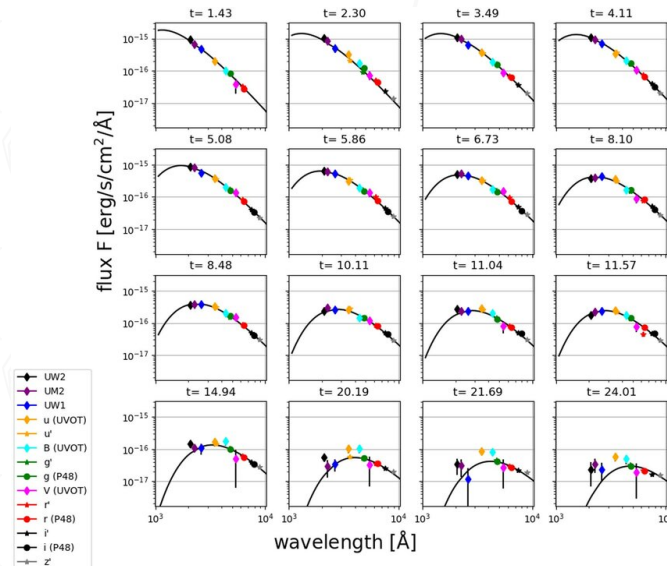


Figure 1: Fits to the multi-epoch photometry of a transient SN 2019hgp using a combined blackbody (black curve) for different epochs observations. You can find the larger version of this figure at: [Link](https://www.nature.com/articles/s41586-021-04155-1/figures/8)

Question 1 (20 marks)

We use observed flux at different filters to calculate the evolution of temperature of an astrophysical object. In Fig. 1 the multi-wavelength data for a core-collapse supernova is presented from Gal-Yam et al. 2022¹. In the figure t is the time in days since the explosion of the supernova.

- (4 marks) We can fit blackbody curve to the observed data to calculate the temperature of the spectrum. Looking at the black lines in Fig. 1, what is an approximate temperature at $t = 1.43$ days since explosion?
- (4 marks) Looking at the different panels, how is the temperature of the supernova changing with time explain in one to two lines? Can you estimate the difference in temperature from the first epoch to last epoch?
- (4 marks) Looking at the data and the curve fit, from what epochs does the blackbody fit deviates from the observed data? What can you say about the reliability of derived temperature value from this epoch onward in one to three sentences?
- (4 marks) How does f_λ at 10^4 Å change with time and what does this imply about the behaviour of the solid angle of the supernova (qualitative answer in one or two sentences suffice)? If you need more information to answer this question then state what this information would be.
- (4 marks) Assuming Rayleigh Jeans approximation at $t = 1.43$ days after the explosion and the distance to SN 2019hgp to be 298 Mpc, calculate the radius of the supernova. Hint: use f_λ at particular wavelength like 10^4 Å and temperature estimate from previous question. An approximation of the value is good for this calculation.

¹<https://www.nature.com/articles/s41586-021-04155-1/figures/8>

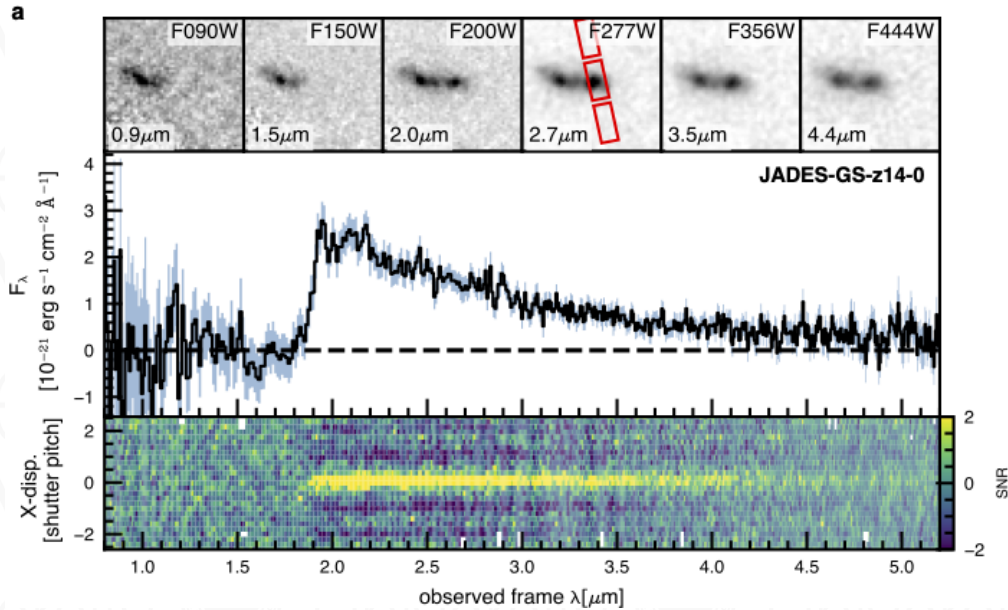


Figure 2: The James Webb Space Telescope spectrum of distant galaxy JADES-GS-z14-0. The figure is also available online at <https://www.nature.com/articles/s41586-024-07860-9/figures/1>.

Question 2 (28 marks)

The 6.5-metre James Webb Space Telescope has obtained spectra of redshift $z \sim 14$ galaxies using the NIRSpec infrared spectrograph which is equipped with a prism and diffraction gratings to achieve a range of spectral resolutions. At $\simeq 3$ micron wavelength ($\nu \simeq 10^{14}$ Hz) the spectral resolution with the prism is $R \simeq 100$ and the linear dispersion is 0.014 micron per pixel.

The preprint of the relevant paper can be found at Paper and a short technical summary of NIRSPEC is available via Link.

- (13 marks) Given the approximate value of f_λ at 3 microns, calculate the flux density in SI units. What is the approximate number of source photons per second per square metre per micron (i.e. photons per unit time per unit area per unit wavelength)?
- (7 marks) What is the collecting area of JWST? The total exposure time on JADES-GS-z14-0 with the NIRSPEC prism was 33,612 seconds. For an idealised case, approximately how many photon counts would you expect per pixel in the NIRSPEC spectrum at 3 microns? (If you struggled to answer the previous part of this question, use a photon count density of 0.1 photons per second per square meter per micron.)
- (4 marks) Given the photon counts per pixel and the information provided here, what would be the maximum expected signal-to-noise to be per pixel? How does this compare with what you can see the observed spectrum? (If you struggled to answer the previous part of this question, use photon counts of 5000 photons.)
- (4 marks) Given James Webb Space Telescope was operating nominally when observing, what are two reasons why the signal-to-noise could be lower than the idealised case?

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Useful information

Physical Constants:

G	gravitational constant	$= 6.673 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2} = 6.673 \times 10^{-8} \text{ cm}^3 \text{ g}^{-1} \text{ s}^{-2}$
$\Re$	universal gas constant	$= 8.314 \times 10^7 \text{ erg K}^{-1} \text{ g}^{-1}$
k	Boltzmann's constant	$= m_u \Re = 1.38 \times 10^{-16} \text{ erg K}^{-1} = 1.38 \times 10^{-23} \text{ J K}^{-1}$
a	radiation density constant	$= \frac{4\sigma}{c} = 7.56 \times 10^{-16} \text{ J m}^{-3} \text{ K}^{-4}$
σ	Stefan-Boltzmann constant	$= 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
c	speed of light	$= 2.998 \times 10^{10} \text{ cm s}^{-1}$
h	Planck's constant	$= 6.62 \times 10^{-34} \text{ J s} = 2\pi\hbar$
m_u	atomic mass unit (amu)	$= 1.66054 \times 10^{-24} \text{ g} \equiv 931.5 \text{ MeV}$
$R_\odot$	solar radius	$= 6.96 \times 10^{10} \text{ cm}$
$M_\odot$	solar mass	$= 1.989 \times 10^{33} \text{ g}$
$L_\odot$	solar luminosity	$= 3.86 \times 10^{33} \text{ erg s}^{-1}$
	1 eV	$= 1.602 \times 10^{-19} \text{ J}$
	1 J	$= 10^7 \text{ erg}$
	1 Jy	$= 10^{-26} \text{ W m}^{-2} \text{ Hz}^{-1}$
	1 AU	$= 1.50 \times 10^{11} \text{ m}$

Physical Formulae:

$$E_\gamma = h\nu = hc/\lambda$$

$$\text{gravitational potential energy: } U = -\frac{GMm}{r}$$

$$\text{blackbody (Planck) function: } B_\nu(T) = \frac{2h\nu^3/c^2}{e^{h\nu/kT} - 1}$$

$$\text{redshift: } z = \frac{\Delta\lambda}{\lambda} = \frac{\lambda_{\text{obs}}}{\lambda_{\text{emitted}}} - 1$$

$$\text{Doppler shift: } \frac{v}{c} = z = \frac{\Delta\lambda}{\lambda_0} \text{ if } v \ll c, \text{ otherwise } \frac{v}{c} = \frac{(1+z)^2 - 1}{1+(1+z)^2}$$

$$\text{ideal gas equation of state: } P = \frac{\rho kT}{\mu m_H}$$

$$\text{gravitational time dilation: } \Delta t_\infty = \Delta t_0 \left(1 - \frac{2GM}{Rc^2}\right)^{-1/2} \text{ \& redshift } \nu_\infty = \nu_0 \left(1 - \frac{2GM}{Rc^2}\right)^{1/2}$$

$$\text{Schwarzschild Radius: } r_S = \frac{2GM}{c^2}$$

Astronomical Formulae:

$$\text{Flux of an object of luminosity } L \text{ (radiating photons isotropically) at distance } d: F = \frac{L}{4\pi d^2}$$

$$\text{Stefan-Boltzmann law for luminosity of an object with radius } R \text{ and temperature } T: L = 4\pi R^2 \sigma T^4$$

$$\text{Stefan-Boltzmann law for surface flux of an object with temperature } T: F = \sigma T^4$$

$$\text{Wien's law (with } T \text{ in units of K): } \lambda_{\text{max}} = 0.002897755 T^{-1} \text{ m or alternatively, } \nu_{\text{max}} = 5.88 \times 10^{10} T \text{ Hz}$$

$$\text{Rayleigh-Jeans approximation: } B_\nu = \frac{2\nu^2 kT}{c^2} = \frac{2kT}{\lambda^2}$$

$$m_1 - m_2 = -2.5 \log_{10} \left(\frac{F_1}{F_2} \right) = -2.5 \log_{10} \left(\frac{B_1}{B_2} \right)$$

$$m - M = 5 \log_{10} \left(\frac{d}{10 \text{ pc}} \right)$$

$$\text{circular speed of a particle orbiting a mass } M \text{ at a distance } r: V = \sqrt{\frac{GM}{r}}$$

$$\text{velocity of the Earth around the Sun: } V_E = 30 \text{ km s}^{-1}$$

$$\text{Binary stars: } P^2 = \frac{4\pi^2 a^3}{G(M_1 + M_2)}$$

Statistics Formulae:

$$\text{Gaussian distribution: } f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[\frac{-(x - \mu)^2}{2\sigma^2} \right]$$

$$\text{Poisson distribution: } P_x = \frac{m^x \exp(-m)}{x!}$$

$$\text{Signal-to-noise ratio: } SNR = \frac{S}{\sqrt{S+B(1+A_S/A_B)}}$$